NO.4 i-stop FUNCTION DOES NOT OPERATE [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id1103q1000900

4	i-stop FUNCTION DOES NOT OPERATE
DESCRIPTION	• i-stop function does not operate when vehicle is stopped.
POSSIBLE CAUSE	• The i-stop (engine stop control) permit conditions are not met due to either of the following causes: — Driver operation — Environment (problem other than with vehicle) — i-stop related part malfunction **Note** • PIDs related to i-stop can be displayed in a list using the M-MDS data monitor diagnostic function. • If the engine is stopped by the i-stop control while the following data monitor items on the M-MDS are being viewed, the data monitor items may no longer be displayed. — BRAKE_SW — BRK_F_P_C — LON_ACCL_C — WSPD_SEN_LF — WSPD_SEN_RF — WSPD_SEN_LR — WSPD_SEN_RR

i-stop (engine-stop control) permit condition

Determination module	Determination item	Determination condition		Applicable PID	Inspection item(s)
		ATX	MTX		
PCM	Vehicle speed	0 km/h {0 mph}	3 km/h {2 mph} or less	WSPD_SEN_LF	• Front ABS wheel-speed sensor (See FRONT ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RF	
				WSPD_SEN_LR	• Rear ABS wheel-speed sensor (See REAR ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RR	
	Brake pedal	Brake pedal is depressed while in D or M position	Not applicable	TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].)
				BPA	• Brake switch (See BRAKE SWITCH INSPECTION.)
	Brake fluid pressure	Brake fluid pressure is 2.3 MPa {23 kgf/cm^2, 334 psi} or more while in D or M position	Not applicable	TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].)
				BRK_F_P_C	• Brake fluid pressure sensor (See BRAKE FLUID PRESSURE SENSOR INSPECTION.)
	Accelerator pedal	Released	←	APP	• APP sensor (See ACCELERATOR PEDAL POSITION (APP) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	Clutch pedal	Not applicable	Released	CPP	• CPP switch (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Clutch stroke sensor (See CLUTCH STROKE SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

Determination module	Determination item	Determination condition		Applicable PID	Inspection item(s)
		ATX	MTX		
PCM	MTX gear position	Not applicable	Neutral	CPP/PNP	• Neutral switch (See NEUTRAL SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	Vehicle conditions	Vehicle is stopped in D position*1	Not applicable	TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].)
	Steering speed	15 deg/s or less	←	STR_ROT_SPD	• EPS control module (See EPS CONTROL MODULE INSPECTION.)
	Steering angle	Within 65° of center to left and right (After EPS control module learns center value)	Not applicable	STR_AB_EST	Not applicable
	Steering torque	1.1 N·m {11 kgf·cm, 9.7 in·lbf} or less	←	STR_TRQ_S_M	• EPS control module (See EPS CONTROL MODULE INSPECTION.)
	i-stop OFF switch	OFF	←	I-Stop_OFF	• i-stop OFF switch (See i-stop OFF SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	Vehicle speed history (After engine started and driven once)	3 km/h {2 mph} or more	4 km/h {2.5 mph} or more	I-Stop_VSP	• PCM (See PCM INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	i-ELOOP bypass mode operation condition	Bypass mode is not operated (BYPS_STAT: Bypass Off)	←	BYPS_STAT	• DC-DC converter (i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) INSPECTION [i-ELOOP].)
	Capacitor (i-ELOOP) voltage	14 V or more	←	DCDC_IN_V	• DC-DC converter (i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) INSPECTION [i-ELOOP].)

Determination module	Determination item	Determination condition		Applicable PID	Inspection item(s)
		ATX	MTX		
PCM	Battery charge condition	65.72 % or more	←	BATT_SOC	• Current sensor (See CURRENT SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Battery (See BATTERY INSPECTION.)
	Battery fluid temperature	0—70 °C {32—158 °F}	←	BATT_TEMP	• Current sensor (See CURRENT SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Battery (See BATTERY INSPECTION.)
	Battery voltage	11.2 V or more	←	BATT_V	• Current sensor (See CURRENT SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Battery (See BATTERY INSPECTION.)
	Power brake unit vacuum	−49.8 kPa {−0.508 kgf/cm^2, −7.22 psi} or less (value calculated by subtracting BARO from BBP)	←	BBP	• Power brake unit vacuum sensor (See POWER BRAKE UNIT VACUUM SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
				BARO	• BARO sensor (See BAROMETRIC PRESSURE (BARO) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	Door (driver's side)	Closed	←	DOOR_D	• Front door latch switch (RH) (R.H.D.) Front door latch switch (LH) (L.H.D.) (See FRONT DOOR LATCH SWITCH INSPECTION.)
	Bonnet	Closed *2	←	HOOD	• Bonnet latch switch (See BONNET LATCH SWITCH INSPECTION.)

Determination module	Determination item	Determination condition		Applicable PID	Inspection item(s)
		ATX	MTX		
PCM	Vehicle inclination angle	Within ±7° when level (LON_ACCL_C: within ±0.121 G)	Not applicable	LON_ACCL_C	• SAS control module (See DTC INSPECTION [STANDARD DEPLOYMENT CONTROL SYSTEM].) (See DTC INSPECTION [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)
	Seat belt (driver's side)	Fastened *3	←	SEAT_B_D	• Driver-side buckle switch (See BUCKLE SWITCH INSPECTION.)
	Battery condition setting	Completed	←	Not applicable	Not applicable
	Steering angle sensor initial setting	Completed	Not applicable	Not applicable	Not applicable
	Engine coolant temperature	35—110 °C {95—230 °F} (some models)/55—110 °C {131—230 °F}	←	ECT	• ECT sensor (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	Intake air temperature	100 °C {212 °F} or less	←	IAT	• IAT sensor No.1 (See INTAKE AIR TEMPERATURE (IAT) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
PCM	Altitude	**European (L.H.D U.K.) specs.** • 1,800 m or less (BARO: 81.5 kPa {0.831 kgf/cm^2, 11.8 psi} or more) **Except for European (L.H.D U.K.) specs.** • 1,500 m or less (BARO: 84.6 kPa {0.863 kgf/cm^2, 12.3 psi} or more)	←	BARO	• BARO sensor (See BAROMETRIC PRESSURE (BARO) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	System condition	i-stop related part DTCs are not recorded	←	Not applicable	Not applicable

Determination module	Determination item	Determination condition		Applicable PID	Inspection item(s)
		ATX	MTX		
Start stop unit	+B (ROOM fuse) input	Yes	←	VPWR_B1	• Start stop unit (See START STOP UNIT INSPECTION.) • Battery (See BATTERY INSPECTION.)
	IG1 input	Yes	←	VPWR_IG1	• Start stop unit (See START STOP UNIT INSPECTION.) • IG1 relay • Battery (See BATTERY INSPECTION.)
	ACC input	Yes	←	VPWR_ACC	• Start stop unit (See START STOP UNIT INSPECTION.) • ACC relay • Battery (See BATTERY INSPECTION.)
	IG1 output	Yes	←	RELAY_IG1	• IG1 relay (See RELAY INSPECTION.)
	ACC output	Yes	←	RELAY_ACC	• ACC relay (See RELAY INSPECTION.)
	Push button	Not pressed	←	SSB_1 SSB_2	• Push button start (See PUSH BUTTON START INSPECTION.)
	Door (all)	Closed	←	DOOR_ALL	• Door latch switch (See FRONT DOOR LATCH SWITCH INSPECTION.) (See REAR DOOR LATCH SWITCH INSPECTION.)
	Trunk (4SD)/Liftgate (5HB)	Closed	←	TRUNK	• Trunk lid latch switch (4SD) (See TRUNK LID LATCH SWITCH INSPECTION.) • Liftgate latch switch (5HB) (See LIFTGATE LATCH SWITCH INSPECTION.)
	Seat belt (driver's side)	Fastened *3	←	SEAT_B_D	• Driver-side buckle switch (See BUCKLE SWITCH INSPECTION.)
Start stop unit	Engine start	Engine was started by operating push button with key inside cabin (engine started first time)	←	Not applicable	Not applicable
	Engine rotation speed	300 rpm or more is maintained after engine started	←	Not applicable	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable	Not applicable

Determination module	Determination item	Determination condition		Applicable PID	Inspection item(s)
		ATX	MTX		
TCM	TCM detection condition	All met (I-Stop_EST: On)	Not applicable	I-Stop_EST	• TCM (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].)
	Panic stop	Brake pedal was not depressed suddenly within 4 s	Not applicable	Not applicable	Not applicable
	TFT	20—117 °C {68—242 °F}	Not applicable	TFT	• TFT sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].)
	Gear position	Not in 2nd gear fixed mode	Not applicable	Not applicable	Not applicable
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable	Not applicable
Climate control unit	Cabin temperature (with full-auto air conditioner system)	Difference between target cabin temperature and actual cabin temperature is a certain value or less (Cabin temperature control by A/C is normally performed)	←	Not applicable	Not applicable
	A/C temperature (with full-auto air conditioner system)	Setting other than MIN *5	←	Not applicable	Not applicable
	Ambient temperature	−10—50 °C {14—122 °F}	←	Not applicable	Not applicable
	Warm-up operation condition (with manual air conditioner system)	Ambient temperature is 10 °C {50 °F} or more or engine coolant temperature is 60 °C {140 °F} or more	←	ECT	• ECT sensor (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	Cold condition (with manual air conditioner system)	Ambient temperature is 20 °C {68 °F} or less or evaporator temperature is specified value or less	←	Not applicable	Not applicable
	MODE dial	Except DEFROSTER	←	Not applicable	Not applicable
	Airflow volume control dial (with full-auto air conditioner system)	OFF *6	←	Not applicable	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable	Not applicable

Determination module	Determination item	Determination condition		Applicable PID	Inspection item(s)
		ATX	MTX		
DSC HU/CM *4	Operation request from TCM	Hill Launch Assist (HLA) control request signal from TCM is received	Not applicable	Not applicable	Not applicable
	Wheel speed (4 wheels)	0 km/h {0 mph}	Not applicable	WSPD_SEN_LF	• Front ABS wheel-speed sensor (See FRONT ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RF	
				WSPD_SEN_LR	• Rear ABS wheel-speed sensor (See REAR ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RR	
	Brake pedal	Brake pedal is depressed	Not applicable	BRAKE_SW	• Brake switch (See BRAKE SWITCH INSPECTION.)
	Brake fluid pressure	Approx. 0.45 MPa {4.6 kgf/cm^2, 65 psi} or more	Not applicable	BRK_F_P_C	• Brake fluid pressure sensor (See BRAKE FLUID PRESSURE SENSOR INSPECTION.)
	Accelerator pedal	Released	Not applicable	APP	• APP sensor (See ACCELERATOR PEDAL POSITION (APP) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable	Not applicable

*1 : After the vehicle is stopped, the engine stops if the gear is shifted to N position. In addition, if the gear is shifted to P position after the vehicle is stopped in D position, the engine stop condition by the i-stop control is maintained.
*2 : If the engine is started with the bonnet open, i-stop is inhibited until the engine is stopped.
*3 : If the seat belt is unfastened while the vehicle is stopped, i-stop is inhibited until a vehicle speed record is input once.
*4 : Even if the DSC HU/CM i-stop permit conditions are not met on a flat road surface and downslope, the i-stop function operates. After the i-stop permit condition met on an up slope (slope of **2% or more**), the i-stop function operates.
*5 : When the A/C set temperature is set to MAX, the PCM may inhibit the i-stop control.
*6 : When all of the other i-stop permit conditions related to the climate control unit are met even if the airflow volume control dial is on, the i-stop control is permitted.

Diagnostic Procedure

Step	Inspection	Results	Action
1	**VERIFY DTCs** • Retrieve the PCM, TCM, DSC HU/CM, climate control unit and start stop unit DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5]) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].) (See DTC INSPECTION [DSC HU/CM].) (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) (See DTC INSPECTION [START STOP UNIT].) • Is a DTC displayed?	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5]) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (CW6A-EL)]) (See DTC TABLE [DSC HU/CM]) (See DTC TABLE [FULL-AUTO AIR CONDITIONER]) (See DTC TABLE [START STOP UNIT])
		No	Go to the next step.

Step	Inspection	Results	Action
2	**VERIFY i-stop (ENGINE-STOP CONTROL) PERMIT CONDITION ITEMS** **Warning** **• To prevent an accident, work with two people when the vehicle is being driven (one person drives vehicle, other person operates M-MDS).** **• Observe legal speed limits when driving the vehicle and perform the procedure in a location where safety can be assured.** • Switch the ignition ON (engine off). • Display the PID related to the i-stop (engine-stop control) permit conditions using the M-MDS. (See PID/DATA MONITOR INSPECTION (i-stop) [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Drive the vehicle. • Verify if the i-stop (engine-stop control) permit conditions of the items which can be verified by the PID are met. • Are the permit conditions of all the items met?	Yes	Go to the next step.
		No	Inspect the applicable items for Inspection item(s).
3	**VERIFY VEHICLE CONDITION RELATED TO i-stop (ENGINE-STOP CONTROL) PERMIT CONDITION** • Verify if all the following vehicle conditions are met. — Engine starts by operating push button with key left in vehicle — Brake pedal not depressed suddenly — Gear position: Not in 2nd gear fixed mode • Are all the conditions met?	Yes	Go to the next step.
		No	Verify the i-stop (engine-stop control) permit condition, then go to Step 11.
4	**PERFORM BATTERY CONDITION INITIAL SETTING (i-stop SETTING)** • Perform the battery condition initial setting (i-stop setting). (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).) • Drive the vehicle to verify the i-stop operation condition. • Is engine-stopping by i-stop operable?	Yes	Go to Step 11.
		No	Go to the next step.
5	**INSPECT AMBIENT TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the ambient temperature sensor. (See AMBIENT TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) (See AMBIENT TEMPERATURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) • Is the ambient temperature sensor normal?	Yes	Go to the next step.
		No	Replace the ambient temperature sensor, then go to Step 11. (See AMBIENT TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See AMBIENT TEMPERATURE SENSOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)
6	**INSPECT EVAPORATOR TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the evaporator temperature sensor. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) • Is the evaporator temperature sensor normal?	Yes	Vehicles with full-auto air conditioner system: • Go to the next step. Vehicles with manual air conditioner system: • Go to Step 10.
		No	Replace the evaporator temperature sensor, then go to Step 11. (See EVAPORATOR TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See EVAPORATOR TEMPERATURE SENSOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)

Step	Inspection	Results	Action
7	**INSPECT CABIN TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the cabin temperature sensor. (See CABIN TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the cabin temperature sensor normal?	Yes	Go to the next step.
		No	Replace the cabin temperature sensor, then go to Step 11. (See CABIN TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
8	**DETERMINE IF MALFUNCTION CAUSED BY AIR MIX ACTUATOR MALFUNCTION** • Inspect the air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the air mix actuator normal?	Yes	Inspect the air mix door and linkage for sticking. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) • If the steering switch is normal: — Go to the next step. • If there is any malfunction: — Repair or replace the malfunctioning location, then go to Step 11.
		No	Replace the air mix actuator, then go to Step 11. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
9	**DETERMINE IF MALFUNCTION CAUSED BY AIRFLOW MODE ACTUATOR MALFUNCTION** • Inspect the airflow mode actuator. (See AIRFLOW MODE ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the airflow mode actuator normal?	Yes	Go to the next step.
		No	Replace the airflow mode actuator, then go to Step 11. (See AIRFLOW MODE ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
10	**INSPECT EPS CONTROL MODULE FOR MALFUNCTION** • Inspect the EPS control module. (See EPS CONTROL MODULE INSPECTION.) • Is the EPS control module normal?	Yes	Perform the following procedures, then go to the next step. 1. Switch the ignition off, and after **2 min or more** have elapsed, switch the ignition ON. 2. Start the engine and drive the vehicle on a straight road for **10 m {33 ft} or more** at a speed of **10 km/h {6.2 mph} or more**. 3. Stop the vehicle with the wheels in the straight-ahead position.
		No	Replace the EPS control module, then go to the next step. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
11	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Has the malfunction symptom been eliminated?	Yes	Complete the symptom troubleshooting. (Explain contents of repair to customer)
		No	Repeat the inspection from Step 1.